
DISCOVER: Online Variance-Guided Data Discovery for Budgeted Multimodal GRPO

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Abstract

Recent advances in reinforcement learning have made verifiable rewards a central post-training paradigm for improving multimodal reasoning. However, methods such as GRPO remain data- and compute-inefficient, often spending rollout and annotation budget on prompts that provide little learning signal. We propose DISCOVER, an online data-discovery framework for budgeted multimodal GRPO that enables vision-language models to dynamically identify prompts that are most useful for the current policy. DISCOVER first organizes the multimodal sample pool into semantic clusters and queries a small set of density-diverse representative samples to obtain initial reward observations. During training, it estimates the utility of unannotated samples by transferring sparse reward feedback from queried samples. Since sample utility changes as the policy evolves, DISCOVER dynamically updates these estimates with recency-weighted rollout observations. It then selects only high-utility prompts for annotation using a policy-conditioned score that accounts for reward variance and difficulty, and continues utility-based replay once the discovery budget is exhausted. Experiments on ViRL39K and VLAA-Thinking with Qwen3-VL-2B show that DISCOVER substantially improves multimodal mathematical reasoning under a strict 10% unique-sample budget, outperforming budget-matched random selection, hard-example sampling, full-pool GRPO under the same training-step budget, and online selection baselines. Ablations further validate the importance of semantic coverage, online reward transfer, and stable discovery cadence, establishing DISCOVER as a practical framework for data-efficient multimodal RL post-training.

1 Introduction

Reinforcement learning with verifiable rewards (RLVR) is the leading post-training paradigm for reasoning in large language and vision-language models (LLMs and LVLMs). A primary algorithm is Group-Relative Policy Optimization (GRPO) [30], which replaces the learned critic of PPO [27] with reward comparisons across G rollouts per prompt. GRPO has significantly improved mathematical and visual reasoning for text-only [7, 20, 42] and vision-language models [12, 36, 43].

The primary computational bottleneck in GRPO is the rollout-and-grade pipeline requiring G stochastic inferences and rewards per prompt. GRPO is inherently inefficient when prompts yield identical semantic outcomes across rollouts, as these produce no policy-gradient signal [39, 42, 46]. Many prompts are either too easy, too difficult, or redundant, suggesting that learning utility varies by prompt and evolves alongside the model. Beyond this computational overhead, uniform sampling also incurs high human costs for verifiable rewards, which often require expert annotation (e.g., from physicians or mathematicians). Consequently, practical applications must limit both computational overhead and the total volume of labeled data.

Existing approaches address these limitations through several strategies. **Static and offline data selection** methods score a fixed pool by quality, gradient features, or embedding geometry and select a subset [9, 18, 21–23, 40, 44]. **Online prompt selection for RL** estimates difficulty or variance from rollouts to skip or up-weight prompts [3, 10, 25, 31, 33, 35, 39, 41, 46]. **Multimodal RL post-training** focuses on adjusting optimization signals, replay strategies, or initializations [12, 36, 43]. However, these methods generally lack *adaptive selection that evolves with the model* or require full human answer annotations.

We introduce an online data-discovery method—DISCOVER—addressing these core limitations. Prior to training, DISCOVER embeds image-question pairs using a multimodal encoder [17], clusters the embedding space, and obtains ground-truth rewards for density-diverse representatives (REPR) of each cluster. During training, it uses cosine-KNN regression to predict the mean and variance of the reward for the full prompt set and subsequently selects training subset via variance-guided utility and cluster coverage constraints. These subsets are then added to a time-decayed buffer. DISCOVER continues to discover new prompts until the annotation budget is exhausted; afterward, it resamples from the frozen pool based on continuously updated utility scores. Thus, the selector functions as a closed-loop controller, allocating the unique-prompt budget and preserving diversity as the policy evolves, while leveraging standard GRPO rollouts as a cost-efficient selection signal.

DISCOVER delivers strong empirical performance, which we demonstrate by training on the ViRL39K [36] and VLAA-Thinking [5] datasets under a strict 10% total verified-prompt budget, counting both GRPO-training prompts and selector-only probes. Our approach yields significant improvements across four multimodal math benchmarks: DynaMath [48], MathVerse [45], MATH-Vision [37], and WeMath [24]. Specifically, for Qwen3-VL-2B, DISCOVER achieves a **+22.5%** relative gain over a 10% random baseline at identical training costs. These results, supported by detailed ablations, highlight the critical role of our data discovery and selection strategy.

Contributions. Our contributions can be summarized as follows:

1. **A policy-conditioned utility score for multimodal GRPO.** We introduce a selection score targeting prompts with high GRPO signal utility. The score integrates predicted reward variance, a mean-aware hard-side bias, and an informative reward band for the generated answer (Section 3.3).
2. **Efficient sparse-to-dense score transfer.** To avoid the costs of dense rollouts and visual probes, DISCOVER measures reward statistics on a small online subset. These are stored in recency-weighted buffers and transferred to the full prompt pool via cosine-KNN regression over frozen multimodal embeddings (Sections 3.3).
3. **Data-efficient multimodal GRPO evidence.** This score-and-transfer strategy improves multimodal mathematical reasoning under a fixed unique-prompt budget. Through ablations, we isolate the roles of coverage, online score transfer, utility shaping, and cadence stability. A held-out audit verifies that the gains also transfer to image-dependent prompts. (Section 4).

2 Related Work

RL post-training and GRPO. GRPO and its variants [30, 7, 20, 42] adapt PPO [27] to verifiable rewards using group-relative normalization instead of a learned critic. Theory suggests GRPO efficiency peaks in a balanced-difficulty regime determined by a relative budget [34]; as the policy (model) improves, the ideal training prompts are non-stationary. DISCOVER tracks this moving target via recency-weighted estimation, dynamically curating the active set under a fixed unique-prompt budget without modifying the GRPO objective.

Online prompt selection for RL post-training. Recent work scores prompts adaptively during RL fine-tuning via reward variance, difficulty filtering [35, 46, 33], learning impact [18], Bayesian posteriors [25, 31], entropy verification [39], or curriculum ranking [3, 10]. Closest to our work, VADE [11] maintains Beta posteriors for observed samples and uses Thompson sampling with prior decay. Under a strict unique-prompt budget, DISCOVER instead transfers recent disagreement and mean estimates through the embedding graph, so unqueried prompts can be ranked before spending verification budget, and couples this transfer with cluster coverage. MMR1-VAS [16] combines outcome variance and trajectory diversity, but the latter requires post-generation data, failing to score never-queried prompts. BOTS [31] employs Bayesian difficulty transfer but is primarily text-focused.

DISCOVER uses embedding transfer and cluster coverage to decide which new prompts should receive verifiable feedback.

Curricula, active learning, and coresets. Curriculum and self-paced learning [4, 15] use easy-to-hard orderings under a fixed score, while active learning and coresets address label-efficient querying [29] and representative subset selection [28, 26]. Batch active learning methods like BADGE [1], BatchBALD [14], and Cluster-Margin [6] jointly optimize diversity and uncertainty within each selected batch for supervised tasks. Unlike these, DISCOVER operates on prompt selection under a rollout budget, deriving uncertainty from live GRPO reward variance and recomputing scores online at no additional cost.

Multimodal RL post-training. VL-Rethinker [36], R1-VL [43], and Vision-R1 [12] each modify the optimizer signal, replay schedule, or initialization for multimodal GRPO, but treat the prompt pool as a fixed, exchangeable set. Recent optimizer-side methods further address perception or stability directly: PAPO adds perception-aware regularization [38], MSSR stabilizes single-rollout multimodal RLVR [19], CalibRL controls exploration with hybrid-policy calibration [13], and RC-GRPO injects reward-conditioned rollout diversity [47]. These methods change how the policy is updated or how rollouts are generated. DISCOVER optimizes scarce feedback allocation and is composable with existing optimizer-side improvements.

3 Method

3.1 Problem Setting and Overall Framework

We introduce DISCOVER, an online data-discovery framework for budgeted multimodal GRPO. To train an effective policy while querying verifiable rewards for only a small fraction of the prompt pool $\mathcal{D} = \{x_i\}_{i=1}^N$ (where $x_i = (q_i, I_i)$), DISCOVER employs a structured pipeline: unsupervised semantic clustering, initialization via representative samples, dynamic utility estimation using recency-weighted cosine-KNN transfer, and cluster-balanced discovery transitioning to frozen-pool replay once the budget is exhausted.

Problem Setting. In the budgeted multimodal GRPO setting, a policy π_{θ_t} is trained at step t . For a sample x_i , GRPO samples G completions:

$$y_{i,1}, \dots, y_{i,G} \sim \pi_{\theta_t}(\cdot \mid x_i), \quad (1)$$

yielding scalar rewards $r_{i,g}^{(t)}$. The empirical group mean and variance are:

$$\mu_i(t) = \frac{1}{G} \sum_{g=1}^G r_{i,g}^{(t)}, \quad v_i(t) = \frac{1}{G} \sum_{g=1}^G \left(r_{i,g}^{(t)} - \mu_i(t) \right)^2. \quad (2)$$

Budgeted goal. Because verifiable feedback is costly, we constrain the set of unique prompts receiving verifier feedback, $\mathcal{V}_{\leq t}$, to a strict budget B :

$$|\mathcal{V}_{\leq t}| \leq B = \lfloor \beta N \rfloor. \quad (3)$$

where β is the budget fraction. Reward variance $v_i(t)$ acts as a policy-dependent utility proxy; samples with uniform rewards (all correct or all incorrect) provide no relative signal. The challenge lies in estimating this time-dependent utility for unqueried prompts while ensuring semantic coverage and keeping estimates aligned with the current policy.

Overall Framework.

The two-stage pipeline is illustrated in Figure 1:

1. **Offline Multimodal Sample Organization:** The pool is partitioned into semantically coherent clusters using spherical k -means on frozen multimodal features. Density-diverse representatives (**REPR**) are selected to seed initial reward observations and provide neighborhoods for utility transfer.
2. **Online Data Discovery and Frozen Reweighting:** During training, DISCOVER uses rollout feedback to estimate sample utility. Sparse feedback is transferred via cosine-KNN and kept current with recency weighting. High-utility samples are admitted through cluster-balanced discovery until B is reached, after which pool is frozen & replay is dynamically reweighted.

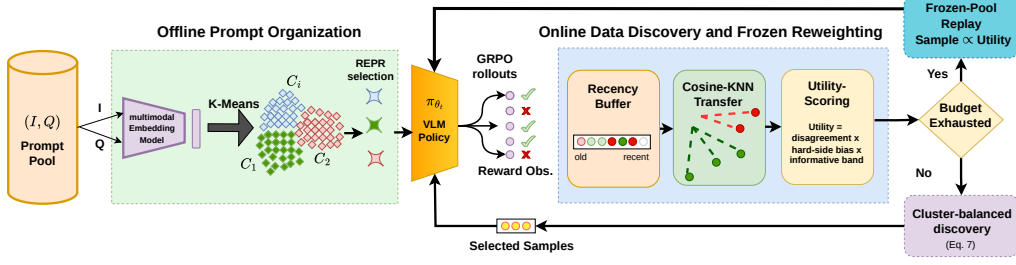


Figure 1: **DISCOVER pipeline.** *Offline* (green, left): prompts are embedded, clustered via spherical k -means, and seeded with REPR samples. *Online* (blue, right): GRPO rollouts from π_{θ_t} feed a recency-weighted buffer; cosine-KNN transfers sparse statistics to unobserved prompts; utility scoring (Eq. 7) ranks all candidates. The budget switch routes scores to cluster-balanced discovery (Eq. 8), which admits new prompts until budget B is exhausted, after which frozen-pool replay (Eq. 9) reweights the existing pool. Both modes return prompts to the policy.

Initialization evaluates $n_{\text{reps}} \times K$ REPR samples, counting toward B . Newly selected samples refresh the recency-weighted buffer, allowing DISCOVER to act as a closed-loop controller for data selection.

3.2 Offline Multimodal Sample Organization

To ensure semantic diversity, DISCOVER extracts L_2 -normalized embeddings $z_i \in \mathbb{R}^d$ ($d = 2048$) using Qwen3-VL-Embedding-2B. These fixed features support both clustering and cosine-KNN transfer. We apply spherical k -means to obtain K clusters, serving as semantic regions for balanced budget allocation.

To seed the observation buffer, we select n_{reps} non-redundant REPR samples per cluster. Candidates are ranked by a density-centrality score—the product of their embedding’s inner product with the cluster center and their local neighborhood density. We then apply greedy selection with a similarity penalty against already chosen representatives to suppress near-duplicates, ensuring REPR samples cover multiple local modes within each cluster.

3.3 Online Data Discovery and Frozen Reweighting

DISCOVER enhances GRPO training through an online discovery loop that estimates reward means and disagreements for all prompts, even when reward observations are sparse. It initializes with representative samples (REPR), maintains statistics via recency weighting, and uses cosine-KNN to infer values for unqueried prompts. After filtering by multimodal utility and selecting new prompts through cluster-balanced discovery, the system freezes the pool once the budget is reached and reweights replay based on the most current utility updates. These steps are explained in the following.

Initialization with REPR samples. Training begins by evaluating $n_{\text{reps}} \times K$ representative samples (selected offline) using G rollouts. This seeds reward observations across diverse semantic regions and initializes the set $S_{\leq 0}$ within the unique-prompt budget B .

Sparse-to-dense transfer with cosine-KNN. To estimate utility for the majority of unqueried prompts, DISCOVER transfers statistics from the observed set R_t using cosine-KNN regression on frozen multimodal features z :

$$\hat{y}_i(t) = \sum_{j \in \text{NN}_k(i; R_t)} \alpha_{ij}(t) y_j, \quad \alpha_{ij}(t) = \frac{\exp(\langle z_i, z_j \rangle / \tau)}{\sum_{\ell \in \text{NN}_k(i; R_t)} \exp(\langle z_i, z_\ell \rangle / \tau)}. \quad (4)$$

With $k = 64$ and $\tau = 0.05$, this nonparametric approach predicts reward means $\hat{\mu}_i(t)$ and disagreement $\hat{v}_i(t)$ without requiring an auxiliary selector network.

Recency-weighted rollout buffer. To track the evolving policy, DISCOVER maintains a buffer where observations at step τ are discounted by $w(t, \tau) = \exp[-\lambda(t - \tau)]$ ($\lambda = 0.05$). Group variance is normalized by the Bernoulli envelope to isolate usable contrast:

$$\tilde{v}_i(\tau) = \text{clip}_{[0,1]} \left(\frac{v_i(\tau)}{\mu_i(\tau)(1 - \mu_i(\tau)) + \epsilon} \right). \quad (5)$$

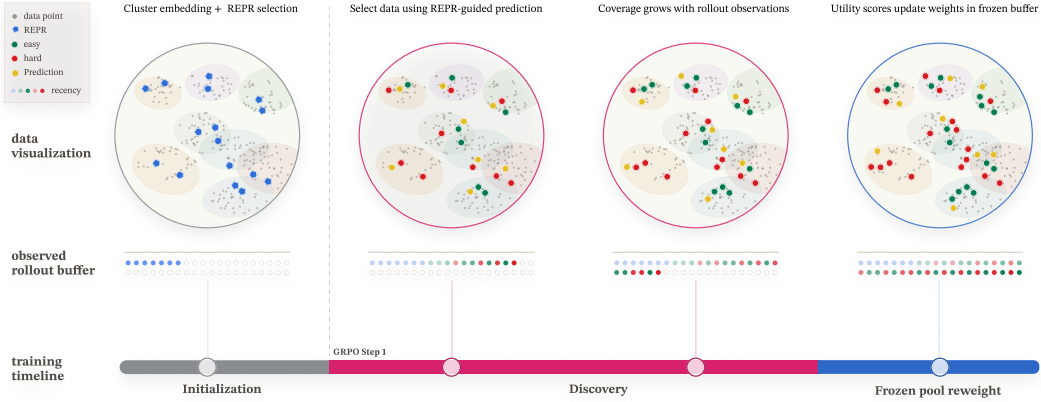


Figure 2: **Cluster-balanced online discovery and frozen-pool reweighting.** DISCOVER builds a fixed multimodal embedding atlas, clusters the prompt pool, and probes REPR to seed a recency-weighted buffer. During discovery, observed easy/hard prompts (green/red) are admitted across clusters while cosine-KNN transfers sparse feedback to unobserved candidates (yellow). Once the unique-prompt budget is exhausted, the pool is frozen and subsequent GRPO steps reweight replay from the buffer; intensity encodes recency.

The resulting recency-weighted statistics, $\bar{\mu}_i(t)$ and $\bar{v}_i(t)$, ensure that stale observations naturally lose influence:

$$\bar{\mu}_i(t) = \frac{1}{Z_i(t)} \sum_{\tau \in \mathcal{T}_i} w(t, \tau) \mu_i(\tau), \quad \bar{v}_i(t) = \frac{1}{Z_i(t)} \sum_{\tau \in \mathcal{T}_i} w(t, \tau) \tilde{v}_i(\tau). \quad (6)$$

Utility-based filtering. DISCOVER identifies high-value prompts using a composite utility that balances disagreement with the model’s current capability frontier:

$$u_i(t) = \underbrace{\hat{v}_i(t)}_{\text{disagreement}} \cdot \underbrace{a(\hat{\mu}_i(t))}_{\text{asymmetry}} \cdot \underbrace{m_{\text{dz}}(\hat{\mu}_i(t))}_{\text{dead zone}}. \quad (7)$$

The asymmetry factor $a(\hat{\mu}_i(t)) = 1 + \eta_h(0.5 - \hat{\mu}_i(t))$ favors harder prompts, while the dead zone filter $m_{\text{dz}}(\hat{\mu}_i(t)) = \mathbb{I}\{\ell \leq \hat{\mu}_i(t) \leq h\}$ excludes solved or impossible tasks.

Cluster-balanced discovery. New prompts are admitted until budget B is reached. DISCOVER allocates the step-wise budget b_t across clusters based on their average utility, selecting the top candidates within each:

$$\mathcal{A}_t = \bigcup_c \text{Top}_{b_c(t)} \{u_i(t) : i \in C_c \setminus S_{\leq t}\}. \quad (8)$$

This ensures semantic coverage while prioritizing the most informative data points.

Frozen-pool reweighting. Once the budget is exhausted, the pool $\mathcal{P}$ is frozen. Training continues by sampling prompts with replacement using a utility-proportional distribution and a starvation floor $\epsilon_f = 0.05$ to keep prompts eligible:

$$p_i(t) \propto \mathbb{I}\{i \in E_t\} \max \left(u_i(t), \epsilon_f \max_{j \in E_t} u_j(t) \right). \quad (9)$$

This allows the replay frequency to adapt dynamically as the policy improves.

4 Experiments

4.1 Setup

Training setup. We fine-tune Qwen3-VL-2B-Instruct [2] with GRPO [30] following standard training protocol from [32] on two multimodal reasoning datasets : ViRL39K [36] and VLAA-Thinking [5]. Unless otherwise stated, all GRPO-trained rows use the same optimizer-step budget,

rollout group size, hardware, and training implementation: 300 steps for ViRL39K and 170 steps for VLAA-Thinking, $G=8$ rollouts per prompt, $4\times\text{H100 GPUs}$, HybridFlow [32], and exact-match rule-based verifiers. The step counts correspond to one pass over the full eligible training pool (i.e. one epoch), or roughly ten passes over a 10% budgeted training pool.

Budget and exposure accounting. We track two cost axes separately. Annotation cost corresponds to unique prompts that receive any verifier feedback, split into **Train** (admitted into GRPO updates) and **Probe** (used only for selector scoring). Rollout cost **Probe Gen.** is reported as a multiple of the source-pool size N : $1N$ for one full-pool single-rollout scan, $8N$ for a full G -rollout scan at $G=8$. All GRPO-trained rows share the same optimizer-step budget, group size, and training implementation, so RL compute is matched up to selector overhead. Budgeted methods differ in which prompts enter the training pool, not in the number of policy updates. The full per-method breakdown is in Appendix E.

Selector configuration. DISCOVER embeds prompts with frozen Qwen3-VL-Embedding-2B [17], using $K=150$, $R=3$ on ViRL39K and $K=200$, $R=2$ on VLAA-Thinking, both chosen from inertia-elbow and silhouette sweeps (Appendix C). We denote schedules as (new-prompt percentage, interval in steps). The headline schedule uses $(1.0, 4)\rightarrow(0.30, 10)\rightarrow(0.15, 16)$ over the 50%, 85%, and 100% budget phases, while the slow control uses $(0.70, 5)\rightarrow(0.25, 15)\rightarrow(0.10, 25)$. The headline configuration combines semantic coverage, KNN reward-statistic transfer, online refresh, and the asymmetric mean-aware utility. In Table 2, one-shot denotes a single initial discovery pass followed by a frozen pool.

Evaluation protocol. We track training stability using a held-out 10% validation split and accuracy and actor entropy. For out-of-distribution generalization, we evaluate checkpoints with VLMEvalKit [8] on four external benchmarks covering distinct multimodal mathematical capabilities: DynaMath [48] for visual perturbation robustness, MathVerse [45] for diagram-grounded reasoning, MATH-Vision [37] for competition-level problems, and WeMath [24] for sub-skill decomposition.

Baselines. We compare DISCOVER against (i) the unmodified Qwen3-VL-2B base model; (ii) full-data GRPO under the same optimizer-step budget; (iii) Random- β , a fixed random subset selected before training; (iv) Hard- β , prompts ranked by base-policy failure rate from a single full-pool rollout; (v) MaxVar- β , prompts ranked by base-policy reward variance from a full-pool G -rollout scan; (vi) DOTS-MM, our multimodal adaptation of the DOTS online difficulty selector [35]; and (vii) VADE [11]. DOTS was introduced for LLM RL fine-tuning and estimates adaptive difficulty from a small reference set; for ViRL39K and VLAA-Thinking, we implement DOTS-MM by using the same vision-language policy and verifier to score reference prompts, and we count those reference prompts inside the verified-prompt budget. VADE is reported both in its original full-pool initialization regime and as a 10% train-cap variant. Component ablations isolate the main selector mechanisms and are reported in Section 4.3.

4.2 Main results

Table 1 reports external benchmark accuracy together with unique-prompt exposure. On ViRL39K, DISCOVER reaches a 13.12 four-benchmark average, a +**26.4%** relative improvement over Random-10% (10.38) and a +**66.9%** improvement over the Qwen3-VL-2B base model (7.86). On VLAA-Thinking, DISCOVER reaches 13.19, +**16.5%** above the corresponding Random-10% (11.32) and all other baselines. Unlike Hard- β and MaxVar- β , which train on 10% of prompts but score the entire pool, DISCOVER avoids full-pool verifier scans and reaches the best overall average with only 1.16% Probe exposure on ViRL39K. Full-data GRPO trails the best budgeted selector under the matched step budget, suggesting that which prompts receive repeated GRPO updates matters more than simply maximizing prompt volume.

4.3 Component ablation

Table 2 ablates ViRL39K components under the same 10% cap. The grouped results support five claims about DISCOVER’s design.

(a) Geometric coverage is necessary but insufficient. Cluster-only selection falls below the Random-10% baseline, indicating that spreading the budget across clusters is ineffective without a

Table 1: External multimodal benchmark accuracy with prompt-exposure and selector-cost accounting. Percentages are normalized by each dataset’s full source pool. **Train** is the percentage of distinct source prompts admitted into GRPO updates, **Probe** is the percentage used only for selector feedback, and **Total=Train+Probe** is the verified-prompt exposure. For strict-budget rows, Total is no larger than the percentage in the method name. Dense-scored baselines scan the eligible 90% training split before choosing a training subset, so they are not verified-exposure matched to strict-budget methods. **Probe Gen.** reports selector-only generation cost as a multiple of the full source-pool size N . Accuracies are percentages; bold denotes the best strict-budget result within each dataset block.

Dataset Method		Verified-prompt exposure ↓			Selector compute ↓	External benchmark accuracy (%) ↑				
		Train (%)	Probe (%)	Total (%)		Probe Gen. ($\times N$)	WeMath	MathVerse	MATH-Vision	DynaMath Avg.
Base	Qwen3-VL-2B	–	–	–	–	–	1.78	8.81	4.61	16.23 7.86
ViRL39K [36]	Random-5%	5.00	0.00	5.00	0	11.26	10.18	5.61	16.72	10.94
	Random-10%	10.00	0.00	10.00	0	10.12	9.66	5.59	16.14	10.38
	Hard-10%	10.00	80.00	90.00	0.90	1.67	9.75	7.24	16.31	8.74
	MaxVar-10%	10.00	80.00	90.00	7.20	10.00	10.71	9.87	17.09	11.04
	DOTS-MM-10% [35]	6.64	3.36	10.00	0.27	3.39	8.93	7.24	16.93	9.12
	VADE original [11]	47.77	42.23	90.00	7.20	0.92	9.59	6.25	16.51	8.32
	VADE-10% [11]	10.00	80.00	90.00	7.20	5.86	9.16	6.25	15.35	9.15
	Full train split	90.00	0.00	90.00	0	6.32	9.92	5.92	16.67	9.71
	DISCOVER-5%	3.84	1.16	5.00	0.09	6.85	10.53	7.56	17.21	10.53
	DISCOVER-10%	8.84	1.16	10.00	0.09	13.45	11.12	7.57	20.32	13.12
VLAA-Thinking [5]	Random-5%	5.00	0.00	5.00	0	11.32	12.16	5.92	20.10	12.38
	Random-10%	10.00	0.00	10.00	0	8.68	11.57	5.26	19.76	11.32
	Hard-10%	10.00	80.00	90.00	0.90	4.25	11.47	5.02	18.74	9.87
	MaxVar-10%	10.00	80.00	90.00	7.20	7.83	12.05	6.19	20.96	11.76
	DOTS-MM-10%	5.40	4.60	10.00	0.37	4.10	10.76	5.72	20.38	10.24
	VADE original	47.25	42.75	90.00	7.20	1.62	11.18	4.84	20.08	9.43
	VADE-10%	10.00	80.00	90.00	7.20	6.52	10.90	4.80	18.92	10.29
	Full train split	90.00	0.00	90.00	0	9.25	11.73	6.25	21.30	12.13
	DISCOVER-5%	3.41	1.59	5.00	0.13	8.24	12.52	6.04	20.11	11.73
	DISCOVER-10%	8.41	1.59	10.00	0.13	13.05	13.45	4.93	21.34	13.19

Table 2: Component ablations on ViRL39K under the 10% GRPO training-prompt cap. Mechanism columns indicate which selector modules are enabled. **Avg.** is the four-benchmark external average, and **Ent.** is final actor-policy entropy. Cadence denotes how often new prompts enter the training pool: default is the main adaptive schedule, slow refreshes more conservatively, and one-shot stops after the initial discovery pass.

Group	Intervention	Tests	Transfer	Utility	Refresh	Cadence	Avg. ↑	Ent.
Reference	Full DISCOVER	Reference setting	KNN	asym	✓	default	13.12	0.20
Predictor	Coverage only	Coverage without utility	×	×	×	one-shot	8.23	0.40
	One-shot KNN	One-shot vs. online KNN	KNN	asym	×	one-shot	10.03	0.30
	MLP transfer	Learned vs. local transfer	MLP	asym	✓	default	11.32	0.23
	Ridge transfer	Linear vs. local transfer	Ridge	asym	✓	default	8.80	0.28
	Ridge active probes	More probes for Ridge	Ridge+AP	asym	✓	default	10.86	0.09
	Random score	Random predictor	random	asym	✓	slow	10.76	0.36
Utility	No hard-side bias	Asymmetric utility bias	KNN	symmetric	✓	default	10.43	0.38
	Dead-zone $h=0.8$	Easy-filter aggressiveness	KNN	$h=0.8$	✓	default	11.12	0.12
Cadence	Slow cadence	Delayed-freeze control	KNN	asym	✓	slow	9.51	0.28

per-prompt utility signal. On curated multimodal corpora, many in-distribution prompts are already saturated under the base policy, and coverage alone cannot filter them out.

(b) Sparse-to-dense transfer makes the budget effective. All transfer controls fall below the full method. Random scoring, Ridge, Ridge with active probes, and MLP transfer each lose accuracy despite using the same budgeted discovery loop. The gain therefore does not come from coverage or online refresh alone, but from transferring sparse rollout feedback through local embedding neighborhoods. Even the learned MLP predictor underperforms cosine-KNN, consistent with the

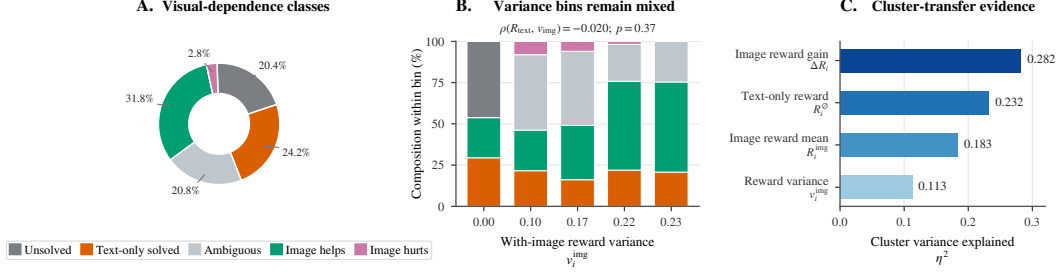


Figure 3: Visual-dependence audit on ViRL39K prompts. (A) Mutually exclusive audit-class breakdown: 24.2% of prompts are solved from text alone. (B) Class composition within bins of with-image reward variance v_i^{img} : variance does not isolate visual dependence ($\rho = -0.020$, $p = 0.37$). (C) Cluster-level variance explained (η^2): embedding clusters explain more variance in image reward gain ΔR_i than in reward variance v_i^{img} (0.282 vs. 0.113).

diagnostics in Section 4.5: the frozen multimodal embedding space already carries useful rank signal, while a parametric head is harder to fit from a small, evolving online buffer.

(c) The pool must refresh as the policy evolves. Immediate freeze runs the KNN ranker only at initialization and then stops admitting new prompts, losing over three points relative to our proposed method. This suggests the round-0 KNN signal does not reliably predict which prompts will be useful later and closed-loop refresh is needed to keep utility estimates aligned with the current policy.

(d) Discovery cadence affects selector quality. Cadence controls how often the selector admits new training prompts. The headline schedule refreshes the pool frequently early in training, then slows admission as the budget fills. The slow control is more conservative, admitting fewer new prompts with longer gaps between refreshes. Its lower performance, despite using the same KNN transfer and utility shape, suggests that early pool growth is important: delaying admission leaves the model training too long on a narrower discovered pool and gives the ranker less useful feedback.

(e) Utility shape contributes a smaller but consistent signal. Removing the asymmetric hard-side bias raises final entropy to 0.38 without late-training collapse, indicating selection has become more diffuse; tightening the dead-zone with $h=0.8$ collapses entropy to 0.12, indicating over-concentration on a narrow band. Both deviations from the default $[0.05, 0.95]$ band degrade the trajectory.

4.4 Visual dependence is statistically independent of reward variance

As a diagnostic, we test whether visual dependence is already captured by reward variance. We audit a random subset of ViRL39K prompts ($n=2,000$) by running $G=8$ text-only rollouts per prompt with the same Qwen3-VL-2B base model. For each prompt, we compare the with-image reward mean R_i^{img} with the text-only reward mean R_i^{t} , record the with-image reward variance v_i^{img} , and define the audit reward gain $\Delta R_i = R_i^{\text{img}} - R_i^{\text{t}}$.

The audit shows that 24.2% of prompts are solved from text alone, while only 31.8% are clearly helped by the image. Moreover, text-only reward and with-image reward variance are uncorrelated (Spearman $\rho = -0.020$, $p=0.37$), indicating that a variance-only selector cannot reliably avoid text-prior shortcuts. Visual dependence is also geometrically structured: cluster identity explains more visual lift than reward variance ($\eta^2=0.282$ vs. 0.113). These results motivate estimating visual dependence separately from reward variance.

Held-out audit probe. We further evaluate DISCOVER on visually grounded prompts of held-out ViRL39K audit fold. DISCOVER improves all audited prompts from 41.7% to 58.7% (+16.9 pp), and improves the image-dependent slice from 62.5% to 79.3% (+16.8 pp). Although the absolute gains are similar across non-visual/mixed (+17.0 pp) and text-leaky (+17.5 pp) slices, the relative error reduction is larger on image-dependent prompts (44.7% vs. 29.0% overall), showing DISCOVER’s gains transfer to visually grounded prompts rather than arising only from text-prior cases.

4.5 Selector diagnostics

Finally, we use selector diagnostics to check whether the online loop behaves as the method assumes. Predictor quality is measured with leave-one-out rank correlation rather than training R^2 , because training fit mostly checks whether recent observations can be memorized. In the slow-cadence

run, leave-one-out Spearman reaches 0.21 for the variance head and 0.39 for the mean head by step 250, indicating that frozen multimodal geometry carries reusable rank signal once the rollout buffer is sufficiently populated. We also log unique-budget usage, frozen-pool size, cluster coverage, cluster-frequency Gini, selection-round Jaccard overlap, selected-versus-unselected predicted utility, entropy, KL, response length, and clipping statistics.

5 Discussion

Interpreting the ablations. The ablations show that budgeted GRPO requires controlling three complementary failure modes. Cluster coverage prevents the budget from collapsing onto a narrow semantic region; online refresh prevents KNN transfer from relying on a stale round-0 policy snapshot; and stable cadence prevents the adaptive ranker from overreacting to outdated rollout-buffer targets. Together, coverage determines where the pool can grow, KNN estimates which prompts are currently useful, and cadence keeps those estimates synchronized with the policy.

Implications for multimodal selection. Static data pruning treats prompt value as fixed, but GRPO changes the value of a prompt as the model learns. This makes budgeted selection a closed-loop control problem rather than a one-shot subset problem. Multimodal data adds a second distinction: a prompt can be useful for GRPO because it induces reward disagreement, yet still be solvable from text priors. The text-only audit shows that reward variance can coexist with text-prior shortcuts, so benchmark gains should be interpreted alongside diagnostic checks for visual dependence.

Scope of transfer. DISCOVER is most useful when samples are heterogeneous, verifiable feedback is expensive, and frozen multimodal embeddings provide meaningful neighborhoods. It is less beneficial when the pool is small enough for exhaustive training, is highly homogeneous, or has embedding neighborhoods that poorly match capability transfer. The text-only audit is corpus-agnostic and can be applied to other multimodal RL datasets to decide whether a visual-dependence channel is needed.

Limitations and future directions. DISCOVER is designed for settings where prompts are heterogeneous, verifier feedback is costly, and frozen multimodal embeddings provide useful neighborhoods for transfer. In this regime, the method adds only a lightweight preparation stage: cached embeddings, semantic clusters, and a small representative-probe set that initializes the online buffer. The present implementation uses fixed choices for the embedding backbone, cluster count, KNN neighborhood size, temperature, recency decay, cadence, and utility thresholds; these choices are intentionally simple and work well in our experiments. A natural next step is to make these controller parameters adaptive, so that the selector can adjust its geometry and discovery rate as the policy changes. More broadly, future versions of budgeted multimodal RL selectors could jointly optimize verified-prompt exposure, rollout cost, semantic coverage, grounding diagnostics, and benchmark-specific transfer within a single multi-objective allocation framework.

6 Conclusion

We introduced DISCOVER, an online data-discovery framework for budgeted multimodal GRPO. Instead of treating the training pool as fixed, DISCOVER spends a limited unique-prompt budget on examples that are predicted to be useful for the current policy: frozen multimodal geometry preserves coverage, recency-weighted rollouts estimate policy-dependent reward statistics, and cosine-KNN transfers sparse feedback to unobserved prompts. The resulting discovery-to-freeze controller builds a compact training pool and then continues to reweight it as the policy changes.

DISCOVER significantly outperforms random and static baselines within the same budget. Ablations show that success requires integrating coverage, online updates, and cadence, as no single ranker is sufficient: : coverage, online refresh, cadence address different failure modes. The text-only audit further reveals that reward variance alone cannot identify visually grounded prompts. These findings suggest treating multimodal RL post-training as dynamic budget allocation under coverage and grounding constraints rather than one-shot data pruning.

References

- [1] Jordan T. Ash, Chicheng Zhang, Akshay Krishnamurthy, John Langford, and Alekh Agarwal. Deep batch active learning by diverse, uncertain gradient lower bounds. In *ICLR*, 2020.
- [2] Shuai Bai, Yuxuan Cai, et al. Qwen3-VL technical report. *arXiv preprint arXiv:2511.21631*, 2025. doi: 10.48550/arXiv.2511.21631.
- [3] Andrei Baroian and Rutger Berger. Prompt replay: Speeding up grpo with on-policy reuse of high-signal prompts. *arXiv preprint arXiv:2603.21177*, 2026.
- [4] Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In *Proceedings of the 26th International Conference on Machine Learning*, pages 41–48, 2009.
- [5] H Chen, H Tu, F Wang, H Liu, X Tang, X Du, Y Zhou, and C Xie. Sft or rl? an early investigation into training rl-like reasoning large vision-language models, 2025. URL <https://arxiv.org/abs/2504.11468>.
- [6] Gui Citovsky, Giulia DeSalvo, Claudio Gentile, Lazaros Karydas, Anand Rajagopalan, Afshin Rostamizadeh, and Sanjiv Kumar. Batch active learning at scale. In *NeurIPS*, 2021.
- [7] DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, et al. Deepseek-rl: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- [8] Haodong Duan, Junming Yang, Yuxuan Qiao, Xinyu Fang, Lin Chen, Yuan Liu, Amit Agarwal, Zhe Chen, Mo Li, Yubo Ma, et al. Vlmevalkit: An open-source toolkit for evaluating large multi-modality models. *arXiv preprint arXiv:2407.11691*, 2024.
- [9] Talfan Evans, Nikhil Parthasarathy, Hamza Merzić, and Olivier J. Hénaff. Data curation via joint example selection further accelerates multimodal learning. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-4485.
- [10] Zhaolin Gao, Joongwon Kim, Wen Sun, Thorsten Joachims, Sid Wang, Richard Yuanzhe Pang, and Liang Tan. Prompt curriculum learning for efficient llm post-training. *arXiv preprint arXiv:2510.01135*, 2025.
- [11] Zengjie Hu, Jiantao Qiu, Tianyi Bai, Haojin Yang, Binhang Yuan, Conghui He, Qi Jing, and Wentao Zhang. VADE: Variance-aware dynamic sampling via online sample-level difficulty estimation for multimodal RL. *arXiv preprint arXiv:2511.18902*, 2025. URL <https://arxiv.org/abs/2511.18902>.
- [12] Wenxuan Huang, Bohan Jia, Zijie Zhai, Shaosheng Cao, Zheyu Ye, Fei Zhao, Zhe Xu, Xu Tang, Yao Hu, and Shaohui Lin. Vision-R1: Incentivizing reasoning capability in multimodal large language models. *arXiv preprint arXiv:2503.06749*, 2025. doi: 10.48550/arXiv.2503.06749. Accepted at ICLR 2026.
- [13] Zhuoxu Huang, Mengxi Jia, Hao Sun, Xuelong Li, and Jungong Han. Controllable exploration in hybrid-policy RLVR for multi-modal reasoning. *International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=5wxyCidRsk>.
- [14] Andreas Kirsch, Joost van Amersfoort, and Yarin Gal. BatchBALD: Efficient and diverse batch acquisition for deep Bayesian active learning. In *NeurIPS*, 2019.
- [15] M. Pawan Kumar, Benjamin Packer, and Daphne Koller. Self-paced learning for latent variable models. In *Advances in Neural Information Processing Systems*, volume 23, 2010.
- [16] Sicong Leng, Jing Wang, Jiayi Li, Hao Zhang, Zhiqiang Hu, Boqiang Zhang, Yuming Jiang, Hang Zhang, Xin Li, Lidong Bing, Deli Zhao, Wei Lu, Yu Rong, Aixin Sun, and Shijian Lu. MMRI: Enhancing multimodal reasoning with variance-aware sampling and open resources. *arXiv preprint arXiv:2509.21268*, 2025. URL <https://arxiv.org/abs/2509.21268>.

- [17] Mingxin Li, Yanzhao Zhang, Dingkun Long, Keqin Chen, Sibao Song, Shuai Bai, Zhibo Yang, Pengjun Xie, An Yang, Dayiheng Liu, Jingren Zhou, and Junyang Lin. Qwen3-vl-embedding and qwen3-vl-reranker: A unified framework for state-of-the-art multimodal retrieval and ranking. *arXiv preprint arXiv:2601.04720*, 2026.
- [18] Xuefeng Li, Haoyang Zou, and Pengfei Liu. Limr: Less is more for rl scaling. *arXiv preprint arXiv:2502.11886*, 2025.
- [19] Rui Liu, Dian Yu, Lei Ke, Haolin Liu, Yujun Zhou, Zhenwen Liang, Haitao Mi, Pratap Tokekar, and Dong Yu. Stable and efficient single-rollout RL for multimodal reasoning. *arXiv preprint arXiv:2512.18215*, 2025. URL <https://arxiv.org/abs/2512.18215>.
- [20] Zichen Liu et al. Understanding rl-zero-like training: A critical perspective. *arXiv preprint arXiv:2503.20783*, 2025.
- [21] Zifan Liu, Amin Karbasi, and Theodoros Rekatsinas. TSDS: Data selection for task-specific model finetuning. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-0325.
- [22] Zikang Liu, Kun Zhou, Wayne Xin Zhao, Dawei Gao, Yaliang Li, and Ji-Rong Wen. Less is more: High-value data selection for visual instruction tuning. *arXiv preprint arXiv:2403.09559*, 2024. doi: 10.48550/arXiv.2403.09559.
- [23] Niklas Muennighoff, Zitong Yang, Weijia Shi, Xiang Lisa Li, Li Fei-Fei, Hannaneh Hajishirzi, Luke Zettlemoyer, Percy Liang, Emmanuel Candès, and Tatsunori Hashimoto. s1: Simple test-time scaling. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 20275–20321, Suzhou, China, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.emnlp-main.1025. URL <https://aclanthology.org/2025.emnlp-main.1025/>.
- [24] Runqi Qiao, Qiuna Tan, Guanting Dong, Minhui Wu, Chong Sun, Xiaoshuai Song, Jiapeng Wang, Zhuoma GongQue, Shanglin Lei, YiFan Zhang, Zhe Wei, Miaoxuan Zhang, Runfeng Qiao, Xiao Zong, Yida Xu, Peiqing Yang, Zhimin Bao, Muxi Diao, Chen Li, and Honggang Zhang. We-Math: Does your large multimodal model achieve human-like mathematical reasoning? In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 20023–20070, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.acl-long.983. URL <https://aclanthology.org/2025.acl-long.983/>.
- [25] Yun Qu, Qi Wang, Yixiu Mao, Vincent Tao Hu, Björn Ommer, and Xiangyang Ji. Can prompt difficulty be online predicted for accelerating RL finetuning of reasoning models? *arXiv preprint arXiv:2507.04632*, 2025. doi: 10.48550/arXiv.2507.04632.
- [26] Akanksha Saran, Safoora Yousefi, Akshay Krishnamurthy, John Langford, and Jordan T. Ash. Streaming active learning with deep neural networks. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 30005–30021. PMLR, 2023. URL <https://proceedings.mlr.press/v202/saran23a.html>.
- [27] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017. doi: 10.48550/arXiv.1707.06347.
- [28] Ozan Sener and Silvio Savarese. Active learning for convolutional neural networks: A core-set approach. In *International Conference on Learning Representations*, 2018.
- [29] Burr Settles. Active learning literature survey. *University of Wisconsin–Madison Computer Sciences Technical Report 1648*, 2009.
- [30] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024. doi: 10.48550/arXiv.2402.03300.

- [31] Qianli Shen, Daoyuan Chen, Yilun Huang, Zhenqing Ling, Yaliang Li, Bolin Ding, and Jingren Zhou. BOTS: A unified framework for bayesian online task selection in LLM reinforcement finetuning. In *International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=rQXBw8fn3Z>.
- [32] Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua Peng, Haibin Lin, and Chuan Wu. Hybridflow: A flexible and efficient rlhf framework. In *Proceedings of the Twentieth European Conference on Computer Systems*, pages 1279–1297, 2025.
- [33] Taiwei Shi, Yiyang Wu, Linxin Song, Tianyi Zhou, and Jieyu Zhao. Efficient reinforcement finetuning via adaptive curriculum learning. *arXiv preprint arXiv:2504.05520*, 2025. doi: 10.48550/arXiv.2504.05520.
- [34] Y. Sun, J. Shen, Y. Wang, T. Chen, Z. Wang, et al. A relative-budget theory for reinforcement learning with verifiable rewards in large language model reasoning. *arXiv preprint arXiv:2602.01523*, 2026.
- [35] Yifan Sun, Jingyan Shen, Yibin Wang, Tianyu Chen, Zhendong Wang, Mingyuan Zhou, and Huan Zhang. Improving data efficiency for LLM reinforcement fine-tuning through difficulty-targeted online data selection and rollout replay. *arXiv preprint arXiv:2506.05316*, 2025. doi: 10.48550/arXiv.2506.05316. Accepted at NeurIPS 2025.
- [36] Haozhe Wang, Chao Qu, Zuming Huang, Wei Chu, Fangzhen Lin, and Wenhui Chen. VL-Rethinker: Incentivizing self-reflection of vision-language models with reinforcement learning. *arXiv preprint arXiv:2504.08837*, 2025. doi: 10.48550/arXiv.2504.08837.
- [37] Ke Wang, Juntong Pan, Weikang Shi, Zimu Lu, Houxing Ren, Aojun Zhou, Mingjie Zhan, and Hongsheng Li. Measuring multimodal mathematical reasoning with MATH-Vision dataset. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-3014.
- [38] Zhenhailong Wang, Xuehang Guo, Sofia Stoica, Haiyang Xu, Hongru Wang, Hyeonjeong Ha, Xiusi Chen, Yangyi Chen, Ming Yan, Fei Huang, and Heng Ji. Perception-aware policy optimization for multimodal reasoning. *arXiv preprint arXiv:2507.06448*, 2025. URL <https://arxiv.org/abs/2507.06448>.
- [39] Jiahao Wu, Ning Lu, Shengcai Liu, Kun Wang, Yanting Yang, Li Qing, and Ke Tang. Train at moving edge: Online-verified prompt selection for efficient RL training of large reasoning model. *arXiv preprint arXiv:2603.25184*, 2026. doi: 10.48550/arXiv.2603.25184.
- [40] Mengzhou Xia, Sathika Malladi, Suchin Gururangan, Sanjeev Arora, and Danqi Chen. LESS: Selecting influential data for targeted instruction tuning. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 54104–54132. PMLR, 2024. URL <https://proceedings.mlr.press/v235/xia24c.html>.
- [41] Yu Yang, Siddhartha Mishra, Jeffrey Chiang, and Baharan Mirzasoleiman. SmallToLarge (S2L): Scalable data selection for fine-tuning large language models by summarizing training trajectories of small models. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-2655.
- [42] Qiying Yu, Zheng Zhang, et al. Dapo: An open-source llm reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476*, 2025.
- [43] Jingyi Zhang, Jiaxing Huang, Huanjin Yao, Shunyu Liu, Xikun Zhang, Shijian Lu, and Dacheng Tao. R1-VL: Learning to reason with multimodal large language models via step-wise group relative policy optimization. *arXiv preprint arXiv:2503.12937*, 2025. doi: 10.48550/arXiv.2503.12937. ICCV 2025 camera ready.
- [44] Jipeng Zhang, Yaxuan Qin, Renjie Pi, Weizhong Zhang, Rui Pan, and Tong Zhang. TAGCOS: Task-agnostic gradient clustered coresets selection for instruction tuning data. *arXiv preprint arXiv:2407.15235*, 2024. doi: 10.48550/arXiv.2407.15235.

- [45] Renrui Zhang, Dongzhi Jiang, Yichi Zhang, Haokun Lin, Ziyu Guo, Pengshuo Qiu, Aojun Zhou, Pan Lu, Kai-Wei Chang, Peng Gao, and Hongsheng Li. MathVerse: Does your multi-modal LLM truly see the diagrams in visual math problems? *arXiv preprint arXiv:2403.14624*, 2024. doi: 10.48550/arXiv.2403.14624.
- [46] Haizhong Zheng, Yang Zhou, Brian R. Bartoldson, Bhavya Kailkhura, Fan Lai, Jiawei Zhao, and Beidi Chen. Act only when it pays: Efficient reinforcement learning for LLM reasoning via selective rollouts. *arXiv preprint arXiv:2506.02177*, 2025. doi: 10.48550/arXiv.2506.02177. Accepted at NeurIPS 2025.
- [47] Haitian Zhong, Jixiu Zhai, Lei Song, Jiang Bian, Qiang Liu, and Tieniu Tan. RC-GRPO: Reward-conditioned group relative policy optimization for multi-turn tool calling agents. *arXiv preprint arXiv:2602.03025*, 2026. URL <https://arxiv.org/abs/2602.03025>.
- [48] Chengke Zou, Xingang Guo, Rui Yang, Junyu Zhang, Bin Hu, and Huan Zhang. DynaMath: A dynamic visual benchmark for evaluating mathematical reasoning robustness of vision language models. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/78b248ea6f627431bba5029d92be8a3d-Abstract-Conference.html.

A Additional visual-dependence proxy evidence

Figure 4 checks whether the audit-derived visual-dependence signal is recoverable from proxies that are available during selection. Both panels use group cross-validation by embedding cluster, so the reported transfer is measured on held-out semantic neighborhoods rather than on prompts from the same cluster. The *No visual signal* label pools prompts that are solved without the image or remain unsolved, since neither provides positive evidence of image dependence.

Table 3 shows that the KNN selectors preserve a positive held-out rank signal across selection rounds, whereas the Ridge logs do not provide a comparable Spearman scalar. Active probing further increases prompt exposure without improving the available held-out diagnostic.

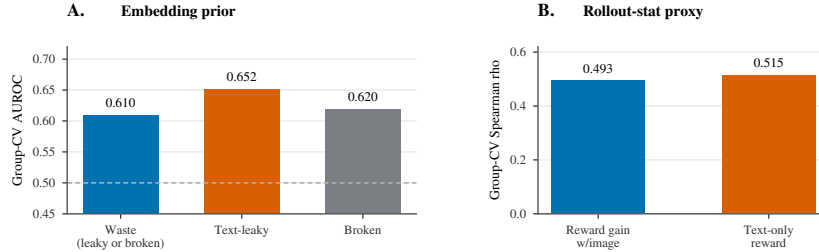


Figure 4: Proxy evidence for visual dependence. (A) The embedding-prior bars report group-CV AUROC for predicting *No visual signal*, *Text-only solved*, and *Unsolved* labels from frozen prompt embeddings, showing above-chance transfer to held-out clusters (AUROC 0.610–0.652). (B) The rollout-stat bars report group-CV Spearman correlation for predicting image reward gain and text-only reward from online rollout statistics, showing moderate transferable rank signal ($\rho=0.493$ and 0.515).

Table 3: Sparse-to-dense transfer diagnostics on ViRL39K. Values are mean $\pm$ standard deviation over all selection steps,

Variant	Selector rule	LOO Spearman
KNN transfer	KNN	0.264 ± 0.063
MLP transfer	MLP	0.182 ± 0.054
Ridge transfer	Ridge	0.142 ± 0.051
Ridge + active probes	Ridge+AP	0.163 ± 0.033
Random-score slow	random	0.103 ± 0.015

Table 4: Selector-cost accounting. N is the eligible training source-pool size, $G=8$ is the rollout group size, and β is the training-prompt budget. Unique-prompt exposure and rollout cost are separated because two methods can verify the same number of unique prompts while requiring different numbers of generations.

Method class	Selector-only feedback	Selector rollout cost	Full-pool scan?	How it appears in Table 1
Random- β	None.	0 selector rollouts.	No	Verified equals Train.
Hard- β	Base-policy failure score for every prompt.	N generations/verifier calls.	Yes	Train = β ; Score-only = $100\% - \beta$; Verified = 100% .
MaxVar- β	Base-policy reward variance for every prompt.	NG generations/verifier calls.	Yes	Same unique exposure as Hard, but $G \times$ higher selector rollout cost.
DOTS-MM	Reference/probe prompts used to train or calibrate the selector.	Method-dependent; not a full-pool verifier scan in our matched run.	No	Reported with actual train uniques and additional score-only probes.
VADE original	Full-pool rollout initialization before its adaptive sampler.	Full-pool selector rollouts.	Yes	Reported at its actual exposure; not treated as budget-matched.
DISCOVER	Sparse representative probes plus rewards already generated by GRPO updates.	No dense verifier scan; non-generative embedding/clustering is excluded.	No	ViRL39K-10%: 8.84% Train, 1.16% Probe, 10.00% Verified.

B Full Selection Algorithm

The ablations in Table 2 correspond to simple switches in this loop. Cluster-only removes the KNN transfer step and allocates by cluster coverage alone. Immediate freeze performs one discovery pass and then sets $z = 1$. Random-score replaces $u_i(t)$ with random scores under the same budget and scheduler. MLP and Ridge replace the nonparametric KNN transfer rule with learned regressors while keeping the remaining discovery loop fixed.

C Embedding and cluster analysis

We build the selector geometry from the ViRL39K train split using 34,983 Qwen3-VL prompt embeddings of dimension 2048. The cached embeddings are already L^2 normalized (mean norm 1.000, standard deviation 1.7×10^{-4}). Figure 5 reports the cluster-count sweep used to choose the operating point. The sweep identifies $K = 150$ as the elbow and recommended value: it keeps the representative-probe set small (449 prompts, 1.16% of the full 38,870-prompt source pool) while preserving semantic coherence. At $K = 150$, the cosine silhouette is 0.089 and the mean dominant-category purity is 0.728; larger K values slightly increase purity but also increase the number of representative probes.

Figure 6 summarizes the selected $K = 150$ geometry. All clusters are nonempty. The mean cluster size is 233.2 prompts, with median 218, 10th–90th percentile range 111.8–390.3, and Gini coefficient 0.248. The embedding spectrum is moderately high-dimensional: the first principal component explains 7.20% of variance, the first 64 components explain 57.5%, the first 128 explain 70.1%, and the participation ratio is 80.8. This supports using a neighborhood-based transfer rule rather than a one-dimensional score or a small number of coarse dataset partitions.

D Bootstrap uncertainty over benchmark items

We estimate evaluation uncertainty by nonparametric bootstrap over benchmark items. For each evaluated model, we resample items within each benchmark, compute that benchmark’s accuracy, and average the four benchmark accuracies; the reported interval is the 2.5–97.5 percentile range

Algorithm 1 DISCOVER: budgeted online discovery for GRPO

Require: Prompt pool $\mathcal{D}$; verifier r ; initial policy π_0 ; rollout group size G ; verified-prompt budget B ; number of clusters K ; representatives per cluster R ; selection schedule $\mathcal{S}$; KNN size k ; decay rate λ .

Ensure: Trained policy π_T ; cap-counted pool $S_{\leq T}$; rollout buffer $\mathcal{H}_T$.

Offline organization

- 1: Embed each prompt $x_i \in \mathcal{D}$ with a frozen multimodal encoder and normalize e_i .
- 2: Partition $\{e_i\}$ into clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$ with spherical k -means.
- 3: Select R density-diverse representatives per cluster to form the REPR set $\mathcal{M}$.

Initialization

- 4: Initialize rollout buffer $\mathcal{H}_0 \leftarrow \emptyset$.
- 5: **for all** $m \in \mathcal{M}$ **do**
- 6: Generate G rollouts with π_0 , verify rewards with r , and append $(m, 0, \{r_{m,g}\}_{g=1}^G)$ to $\mathcal{H}_0$.
- 7: **end for**
- 8: Initialize cap-counted pool $S_{\leq 0} \leftarrow \mathcal{M}$, frozen flag $z \leftarrow 0$, policy $\pi \leftarrow \pi_0$, and replay weights q_0 uniform over $S_{\leq 0}$.

Online discovery and frozen reweighting

- 9: **for** $t = 1, \dots, T$ **do**
 - 10: $S_{\leq t} \leftarrow S_{\leq t-1}$, $\mathcal{H}_t \leftarrow \mathcal{H}_{t-1}$, and $q_t \leftarrow q_{t-1}$.
 - 11: **if** t is a selection step under schedule $\mathcal{S}$ **then**
 - 12: Compute recency-weighted observed means and disagreements $(\bar{\mu}_i(t), \bar{v}_i(t))$ from $\mathcal{H}_t$ using weights $\exp[-\lambda(t - \tau)]$.
 - 13: Let R_t be the prompts with valid recency-weighted statistics.
 - 14: Transfer sparse targets to the full pool with cosine-KNN: $(\hat{\mu}_i(t), \hat{v}_i(t)) \leftarrow \text{KNNTRANSFER}(e_i, \{(\bar{\mu}_j(t), \bar{v}_j(t)) : j \in R_t\}, k)$ for all $i \in \mathcal{D}$.
 - 15: Compute utility
$$u_i(t) = \hat{v}_i(t) a(\hat{\mu}_i(t)) m_{\text{dz}}(\hat{\mu}_i(t)).$$
 - 16: **if** $z = 0$ **then**
 - 17: Mask already admitted prompts: $M_t \leftarrow \mathcal{D} \setminus S_{\leq t}$.
 - 18: Let b_t be the scheduled number of new prompts, clipped so that $|S_{\leq t}| + b_t \leq B$.
 - 19: Allocate b_t across clusters according to each cluster's average utility over M_t .
 - 20: Add the highest-utility allocated candidates from each cluster to $S_{\leq t}$.
 - 21: **if** $|S_{\leq t}| = B$ **then**
 - 22: $z \leftarrow 1$.
 - 23: **end if**
 - 24: **end if**
 - 25: **if** $z = 1$ **then**
 - 26: Update replay weights q_t over the frozen pool $S_{\leq t}$ using $u_i(t)$ and the starvation floor.
 - 27: **end if**
 - 28: **end if**
 - 29: Sample a GRPO batch from $S_{\leq t}$ using q_t .
 - 30: Generate G rollouts per prompt, verify rewards, update π with GRPO, and append the new rewards to $\mathcal{H}_t$.
 - 31: **end for**
 - 32: **return** $\pi_T, S_{\leq T}, \mathcal{H}_T$.
-

over 5,000 bootstrap replicates. These intervals quantify evaluation-set uncertainty for the evaluated models. Because the models are evaluated on the same benchmark items, we also report paired-bootstrap intervals for DISCOVER's gain over each evaluated baseline.

E Prompt-exposure and selector-cost accounting

We report two cost axes: verified-prompt exposure and selector-generation cost. Verified-prompt exposure counts unique prompts that receive verifier feedback, including prompts admitted into GRPO updates (Train) and selector-only probes (Probe). Probe Gen. reports selector-only generation cost as a multiple of the full source-pool size N .

Because 10% of each source pool is held out for validation, dense-scored baselines scan the eligible 90% training split before choosing a 10% training subset. Thus, Hard- β , MaxVar- β , and VADE-10% use 10% Train exposure and 80% selector-only Probe exposure, for 90% total verified exposure. These methods are useful static or online-selection references, but they are not verified-exposure matched to strict-budget methods.

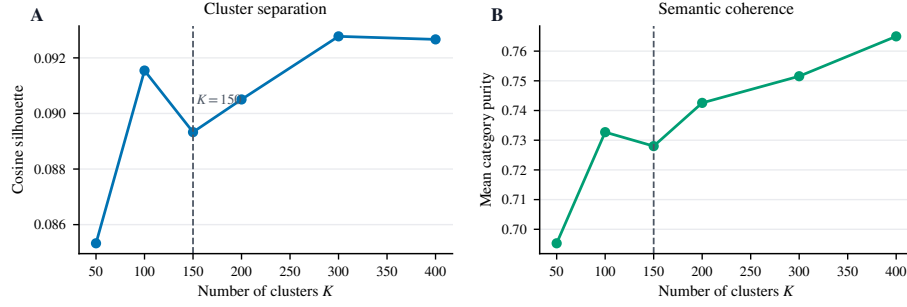


Figure 5: ViRL39K cluster-count sweep on frozen Qwen3-VL embeddings. The dashed line marks the selected operating point $K = 150$. Cosine silhouette and dominant-category purity both show that the embedding geometry groups prompts into coherent neighborhoods, while $K = 150$ keeps representative-probe overhead small.

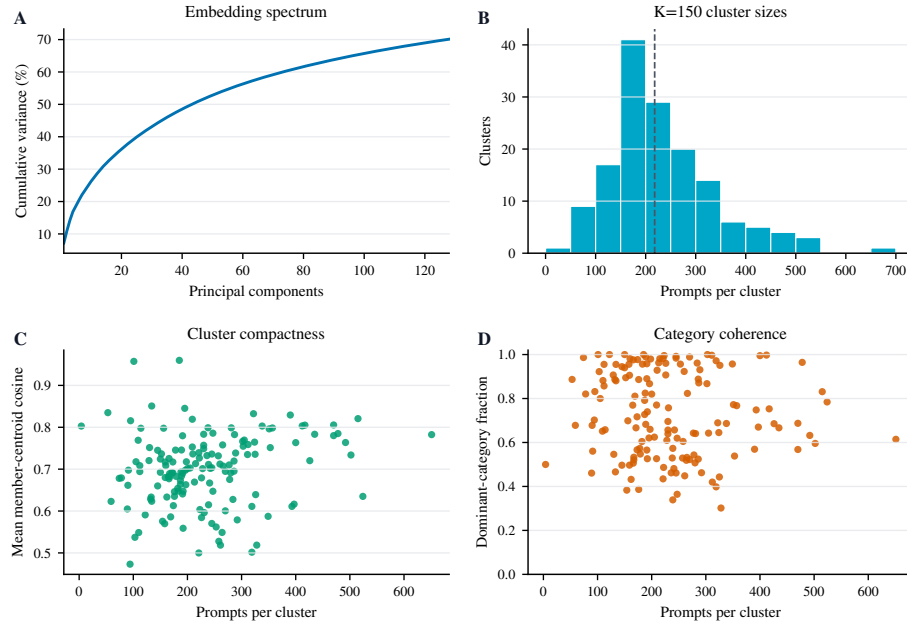


Figure 6: Embedding and cluster statistics for the selected ViRL39K geometry. (A) The frozen embedding spectrum is broad rather than dominated by one direction. (B) The $K = 150$ partition gives moderately balanced nonempty clusters. (C) Member-centroid cosine values show compact semantic neighborhoods. (D) Dominant-category fractions show that many clusters align with dataset-level semantic categories.

Strict-budget methods count both training prompts and selector-only probes toward the named budget. For example, DISCOVER-10% on ViRL39K uses 8.84% Train and 1.16% Probe exposure, for exactly 10.00% total verified exposure. DISCOVER avoids dense verifier scans: its selector overhead consists of representative probes plus non-generative embedding, clustering, KNN transfer, and cluster allocation.

Table 5: Prompt-exposure and selector-cost accounting. N_{all} is the full source-pool size and $N_{\text{tr}} = 0.9N_{\text{all}}$ is the eligible training split. $G=8$ is the rollout group size and β is the total verified-prompt budget, normalized by N_{all} . **Train** denotes prompts admitted into GRPO updates, while **Probe** denotes selector-only prompts. For strict-budget methods, Train+Probe is counted against the same cap. $E(N)$ is one frozen multimodal embedding pass, and $C(N, K)$ is FAISS spherical k -means over the resulting $N \times 2048$ embedding matrix. ViRL39K values use $N_{\text{all}}=38,870$ and $N_{\text{tr}}=34,983$.

Method class	Cap-counted verified exposure	Total touched / selector rollouts	Full-pool rollout scan?	Fairness note
Full-data GRPO	90.00% train, 0.00% probe; total 90.00%	Same prompts as training	No	Fixed-step reference, not an upper bound; it sees more unique prompts but fewer passes per prompt.
Random- β	βN_{all} train, 0 probe; total βN_{all}	Same prompts as training; no selector rollouts	No	Strict budget-matched static subset with no selector overhead.
Hard- β	βN_{all} train and $(0.9 - \beta)N_{\text{all}}$ scoring-only prompts. For $\beta=0.10$: 10.00% train, 80.00% probe; total 90.00%	One full train-split one-rollout scan: N_{tr} generations. ViRL: 34,983 generations, or $0.90 N_{\text{all}}$	Yes	Trains on a β -sized subset but scores the eligible train split, so it is not verified-exposure matched to strict-budget methods.
MaxVar- β	βN_{all} train and $(0.9 - \beta)N_{\text{all}}$ scoring-only prompts. For $\beta=0.10$: 10.00% train, 80.00% probe; total 90.00%	One full train-split G -rollout scan: GN_{tr} generations and rewards. ViRL: 279,864 generations, or $7.20 N_{\text{all}}$	Yes	Strong static variance baseline with dense verifier exposure; not verified-exposure matched.
DOTS-MM-10%	6.64% train, 3.36% probe; total 10.00%	3.36% reference probes plus train prompts, for 10.00% verified exposure	No	Strict verified-budget online selector baseline; DOTS logs do not provide DISCOVER’s LOO KNN or cluster diagnostics.
VADE original	47.77% train, 42.23% initialization/scoring exposure; total 90.00%	One full train-split G -rollout initialization scan: GN_{tr} generations and rewards. ViRL: 279,864 generations, or $7.20 N_{\text{all}}$	Yes	Included as an online-sampling reference with its actual exposure, not as a budget-matched baseline. The percentages convert the original 53.08%/46.92% train-split accounting to the full-source denominator.
VADE-10%	10.00% train, 80.00% initialization/scoring exposure; total 90.00%	One full train-split G -rollout initialization scan: GN_{tr} generations and rewards. ViRL: 279,864 generations, or $7.20 N_{\text{all}}$	Yes	Trains on a 10% subset but requires dense train-split verifier exposure for scoring.
DISCOVER-10%	8.84% train, 1.16% probe; total 10.00%	450 representative probes plus train prompts, for 10.00% verified exposure; $450G = 3,600$ selector-only generations, or $0.09 N_{\text{all}}$	No	Strict verified-budget method. It replaces dense rollout scans with $E(N_{\text{tr}}) + C(N_{\text{tr}}, K)$ non-generative precomputation, in-memory KNN transfer, and cluster allocation. VLL adds at most MS paired log-prob evaluations when enabled.

Table 6: Bootstrap uncertainty over external benchmark items for evaluated ViRL39K models. Intervals are 95% nonparametric CIs for the four-benchmark average.

Method	Avg. $\uparrow$	95% CI
DISCOVER-10%	13.12	[12.21, 14.07]
MLP transfer	11.32	[10.46, 12.27]
Random score	10.76	[9.96, 11.62]
One-shot KNN	10.03	[9.20, 10.93]
Slow cadence	9.51	[8.73, 10.32]
DOTS-MM-10%	9.12	[8.32, 10.00]
VADE original	8.32	[7.58, 9.13]
Coverage only	8.23	[7.46, 9.06]

Table 7: Paired bootstrap gains over evaluated ViRL39K baselines. Each interval resamples the same benchmark items for DISCOVER and the baseline, then recomputes the four-benchmark average difference in percentage points.

Comparison	Δ Avg. (pp) $\uparrow$	95% CI
DISCOVER-10% vs. MLP transfer	1.80	[0.85, 2.73]
DISCOVER-10% vs. Random score	2.35	[1.41, 3.33]
DISCOVER-10% vs. One-shot KNN	3.08	[2.11, 4.06]
DISCOVER-10% vs. Slow cadence	3.61	[2.81, 4.42]
DISCOVER-10% vs. DOTS-MM-10%	3.99	[3.14, 4.83]
DISCOVER-10% vs. VADE original	4.79	[3.96, 5.66]
DISCOVER-10% vs. Coverage only	4.88	[4.07, 5.75]